

Addressing Overfitting

When a model is identified as being **overfit** (having high variance), there are several strategies you can employ to improve its ability to generalize to new data.

1. Collect More Training Data

- **How it works:** Providing the algorithm with more examples helps it learn the true underlying patterns in the data, rather than fitting to the noise in a small dataset. A larger dataset can naturally smooth out a complex, "wiggly" curve.
 - **Effectiveness:** This is often the most reliable way to combat overfitting.
 - **Limitation:** It's not always possible or practical to acquire more data.
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2. Use Fewer Features (Feature Selection)

- **How it works:** Overfitting can be caused by having too many input features relative to the amount of data. By selecting a subset of the most relevant features, you can reduce the model's complexity.
 - **Methods:**
 - **Manual Selection:** Using your domain knowledge and intuition to choose the most important features.
 - **Automated Selection:** Using algorithms (discussed in later courses) to automatically identify the most useful features.
 - **Disadvantage:** You risk **throwing away useful information** by discarding features that might have some predictive value.
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3. Regularization

- **How it works:** This technique keeps all the features but reduces the magnitude (size) of the parameters $\vec{w}$.
- **Intuition:** Overfitting often corresponds to very large parameter values, which create highly complex and sensitive models. Regularization adds a penalty to the cost function for large parameters, encouraging the learning algorithm to find smaller values for them.
- **Effect:** This results in a simpler, smoother model that is less likely to overfit. It's a "gentler" approach than eliminating features entirely.
- **Convention:** Regularization is typically applied to the weight parameters ($w_1, w_2, \dots, w_n$) but not to the bias parameter (b).